Project Title: Apply Community Detection to Find Habitat Clusters

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1. Introduction

Community detection is a technique from graph theory used to partition a network into groups (communities) where nodes are more densely connected to each other than to the rest of the network. This report explores its application in **wildlife conservation** to identify **habitat clusters**. By modeling a landscape as a network—where habitat patches are *nodes* and potential wildlife movement routes are *edges*—community detection algorithms can automatically delineate core habitat areas that function as interconnected units. This helps prioritize conservation efforts by identifying key areas for species preservation and connectivity.

2. Objectives

- To model a geographical area's habitat patches and corridors as a network graph.
- To apply and compare various **community detection algorithms** (e.g., Louvain, Girvan-Newman) to the habitat network.
- To delineate and visualize distinct, interconnected **habitat clusters** or "conservation communities."
- To evaluate how the identified clusters align with known species distribution and conservation needs.

3. Methodology

3.1 Data Preparation and Graph Construction

- **Data Sources:** High-resolution satellite imagery, land-cover maps, and species-specific movement data will be used to define habitat quality.

- **Node Definition:** Core habitat patches (e.g., forest fragments, protected areas) are defined as the network's **nodes**.
- **Edge Definition:** Corridors or resistance surfaces are used to calculate the **edges** (weighted links) between habitat patches, where the weight reflects the ease or probability of animal movement.

3.2 Community Detection Algorithms

1. **Louvain Method:** Used for its efficiency in detecting communities in large networks, which is suitable for complex landscapes.
2. **Infomap:** A flow-based method that focuses on how information (or in this case, a moving animal) flows through the network to define clusters.
3. **Optimization:** The chosen algorithm will be run to maximize the *modularity* score, a metric that quantifies the strength of the division of a network into modules (or communities).

4. Expected Results

- **Clear Habitat Clusters:** The community detection algorithm is expected to output a set of geographically distinct and highly interconnected habitat clusters.
- **Identification of Key Corridors:** Edges with high betweenness centrality that connect different communities will be identified as **critical corridors** that maintain functional connectivity.
- **Prioritization:** The clusters will help conservation planners prioritize efforts by showing which habitat groups are most critical for maintaining a robust, connected network for the target wildlife species.

5. Conclusion

Applying community detection to a habitat network provides a powerful, objective, and data-driven method for defining functional conservation units. By shifting the focus from individual habitat patches to interconnected habitat **clusters**, it supports more effective, landscape-scale conservation planning, directly advancing the goal of AI-Enhanced Wildlife Corridor Identification and Prediction.